Vatsal Maniar

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Education

Stevens Institute of Technology — Hoboken, NJ
M.S. Financial Engineering | GPA: 3.67/4.00 | Graduate Certificate: Financial Risk Engineering — *Sep 2025 – Dec 2026*
Coursework: *Market Microstructure, Stochastic Calculus, Derivatives Pricing & Hedging, Advanced Probability and Statistics*

Arizona State University — Tempe, AZ
B.S. Computer Science | GPA: 3.74/4.00 | Magna Cum Laude — *Aug 2021 – May 2025*

Research

Stock–Bond Diversification in an Inflation-Driven Rates Regime: Regime-Aware Allocation and Treasury Curve Loss Attribution, 2007–2026
SSRN Working Paper / Preprint, 2026 | SSRN
- Evaluated a 13-ETF regime-aware allocation against Equal Weight, GMV, Max Sharpe, SPY, and 60/40, reporting a 0.48 Sharpe versus 0.66 for 60/40 rather than optimizing away the non-dominant result.
- Attributed a standardized long-duration Treasury book's 2022 curve-replay loss of $191.6K across real-rate, breakeven, curve-shape, convexity, and residual channels; ongoing FE800 research extends the framework across historical stress episodes.

Projects

DiffusionWorks: Stochastic Derivatives Numerics | diffusionworks.netlify.app | *C++20, CMake, GoogleTest, QuantLib*
- Engineered C++20 Black-Scholes and Heston derivatives engine spanning analytic, Monte Carlo, finite-difference, and characteristic-function pricing methods across European, Asian, and barrier options.
- Validated pricing and Greek accuracy through 15 formal experiments (10 clean, 5 flagged, zero failed) and 793 automated tests across a 14-job continuous-integration matrix.

Systematic Alpha Research Lab | alpha-research.netlify.app | *Python, pandas, NumPy, SciPy, pytest*
- Designed no-lookahead ETF alpha pipeline across 15 assets and 5 signal families (momentum, reversal, carry, value, quality) with walk-forward validation, 4 cost tiers, and 3 crisis regimes.
- Identified cross-sectional momentum as strongest surviving signal: 8.8% net CAGR, 0.62 Sharpe, and −18% max drawdown after 5 bps costs, outperforming lower-Sharpe reversal evidence.

Event-Market Microstructure & Trading Simulation | apexgp-markets.netlify.app | *Python, FastF1, Recharts*
- Programmed 500+ contracts event-market simulator across 24 Formula One races, updating prices every second from race-state changes, order-flow signals, and futures repricing logic.
- Tracked live-style trading risk for $100K bankroll, including mark-to-market P&L, exposure limits, Sharpe/drawdown, and timestamped transaction audit trails at second-level repricing.

FX Options Risk Lab | fxrisklab.netlify.app | *Python, QuantLib, NumPy, SciPy*
- Modeled USD/MXN option prices and Greeks under Garman-Kohlhagen, reconciling call/put outputs against QuantLib across 11×7 spot-volatility grid.
- Estimated 99% VaR and Expected Shortfall from 10,000 joint spot/volatility Monte Carlo paths, decomposing scenario P&L into Greek exposure and residual risk.

Experience

Alcove Ridge Consulting / Private-Markets.io — Sep 2024 – Jun 2025
Software & Platform Development Intern — *Remote*
- Reduced reporting-preparation time by 30–40% by converting 4 private-markets workflows into sponsor-reviewable dashboards for deal review, fund administration, performance, and oversight.
- Engineered role-based data-access flows for 4 user groups, enforcing consistent permissioning and process control across investor, sponsor, broker, and administrator views.

ASU Capital Assets Management Office — May 2024 – May 2025
Asset Management Specialist — *Tempe, AZ*
- Validated $1.5B in institutional asset records using Excel reconciliation checks and system exports, maintaining 99.9% data accuracy across 4 classification categories.
- Flagged documentation and classification gaps across 1,000+ high-value assets above $5K threshold, improving asset-record integrity for internal-control and sponsor-funded reviews.

Technical Skills

Programming Languages: Python (pandas, NumPy, SciPy, scikit-learn, Matplotlib), C++, R, SQL
Trading Research & Backtesting: Signal Design, Statistical Modeling, No-Lookahead Backtesting, Walk-Forward Validation, Transaction-Cost Modeling, Turnover Analysis, Market Microstructure, Portfolio Construction, Regime/Crisis Stress Testing
Quantitative Finance & Risk: Derivatives Pricing (Black-Scholes, Garman-Kohlhagen), Greeks, Monte Carlo Simulation, VaR/CVaR, Expected Shortfall, P&L Attribution, Scenario Analysis
Platforms & Tools: Git/GitHub, pytest, QuantLib, Bloomberg Terminal, LSEG Workspace, Capital IQ, Excel/VBA